EXPERIMENT 2: STUDY OF CHARGING AND DISCHARGING IN A CAPACITOR AND INDUCTOR

INTRODUCTION

Capacitor charging

Capacitor is said to be in charging mode whenever it is connected to a voltage supply. The capacitor in the circuit is being charged from the power supply with the current passing through the resistor. The initial current, which is the maximum current will be flowing.

$$Initial\ charging\ current, I_o = I_{max} = \frac{E}{R}$$

The current will decrease exponentially until the full state is achieve, while the voltage across the capacitor will increase. This state will continue until the capacitor is fully charged. The current will decrease to zero, while the voltage will increase until the maximum value, which is equal to the supply, E.

At fully charged, Vc = E and I = 0.

The instantaneous capacitor voltage can be represented in an equation as follows. Figure 2 show both curves of current and voltage for the capacitor during the charging process.

$$V_c(t) = V_{max}(1 - e^{-\frac{t}{\tau}})$$

Where $V_c(t) = capacitor\ voltage\ at\ time, t$

$V_{max} = maximum\ voltage = supply\ voltage\ time$

$time\ constant, \tau = RC$

Time constant is the time taken for the capacitor voltage to achieve a value of $0.632V_{max}$ and the current to achieve a value of $0.371I_{max}$. The capacitor is fully charged when the time reaches 5τ.

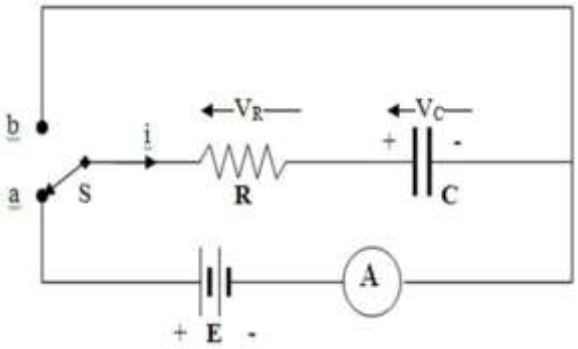


Figure 1: RC Charging circuit

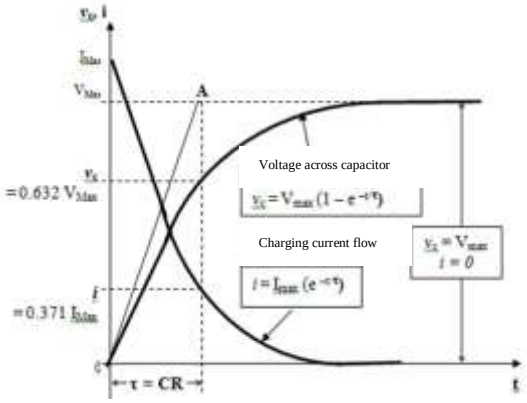


Figure 2

Time Constant	RC Value	Percentage of Maximum	
		Voltage	Current
0.5 time constant	0.5T = 0.5RC	39.3%	60.7%
0.7 time constant	0.7T = 0.7RC	50.3%	49.7%
1.0 time constant	1T = 1RC	63.2%	36.8%
2.0 time constants	2T = 2RC	86.5%	13.5%
3.0 time constants	3T = 3RC	95.0%	5.0%
4.0 time constants	4T = 4RC	98.2%	1.8%
5.0 time constants	5T = 5RC	99.3%	0.7%

Time Constant	RC Value	Percentage of Maximum	
		Voltage	Current
0.5 time constant	0.5T = 0.5RC	60.7%	39.3%
0.7 time constant	0.7T = 0.7RC	49.7%	50.3%
1.0 time constant	1T = 1RC	36.6%	63.4%
2.0 time constants	2T = 2RC	13.5%	86.5%
3.0 time constants	3T = 3RC	5.0%	95.0%
4.0 time constants	4T = 4RC	1.8%	98.2%
5.0 time constants	5T = 5RC	0.7%	99.3%

Figure 3: Percentage of charging and discharging capacitor percentage

Capacitor discharging

By referring to figure 1, when the switch is changed to position b, the discharging process for the capacitor will begin. Figure 3 shows the curve for the discharging process in the capacitor. The capacitor voltage will decrease exponentially during the discharging process as shown in the following equation.

$$V_c(t) = V_{max}(e^{-\frac{t}{\tau}})$$

While the current will increase by

$$I_c(t) = -I_{max}(e^{-\frac{t}{\tau}})$$

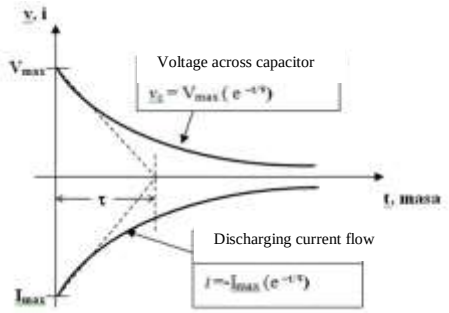


Figure 4

OBJECTIVE

1. To construct charging and discharging capacitor and inductor circuit.
2. To obtained charging and discharging waveform of capacitor and inductor circuit.
3. To measured time constant, current and voltage in charging and discharging waveform.

PART A: CHARGING DISCHARGING OF CAPACITOR CIRCUIT

EQUIPMENT AND COMPONENT

1. Step Voltage, 10V
2. Resistor, 1k Ω
3. Capacitor, 10μ F
4. Switch

PROCEDURE

Circuit diagram

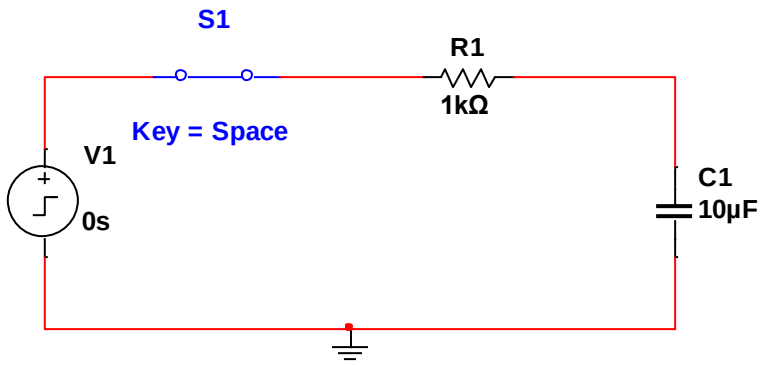


Figure 5

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM106_Experiment 2_YOUR ID_YOURNAME > Save.
3. Wire up the circuit as shown in Figure 5 where R = 1k Ω and C = 10μ F.
4. Set step input voltage to 10V. Initial level = 0V > Final level =10V > step time = 0s > output rise/fall time = 0.1n s.

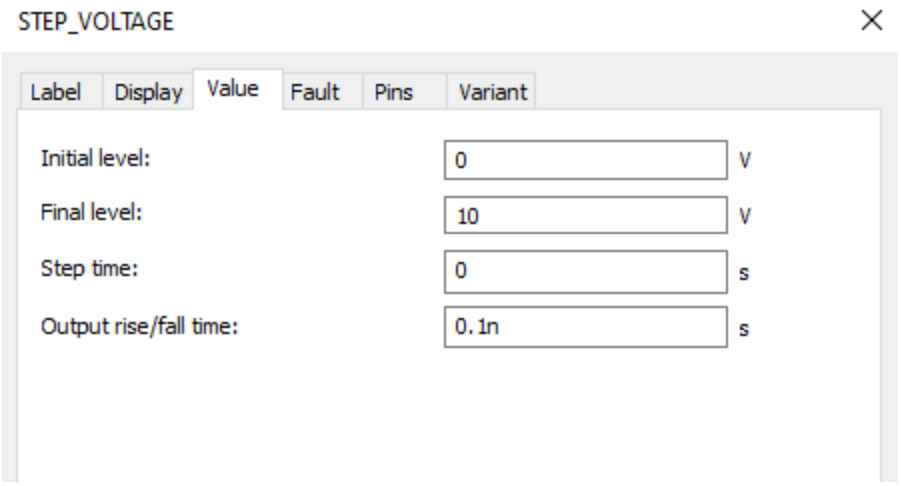


Figure 6

5. Set Analyses and simulation > Transient > Initial condition = set to zero > start time = 0s > end time = 0.1s > maximum time step = 1e-06 s.

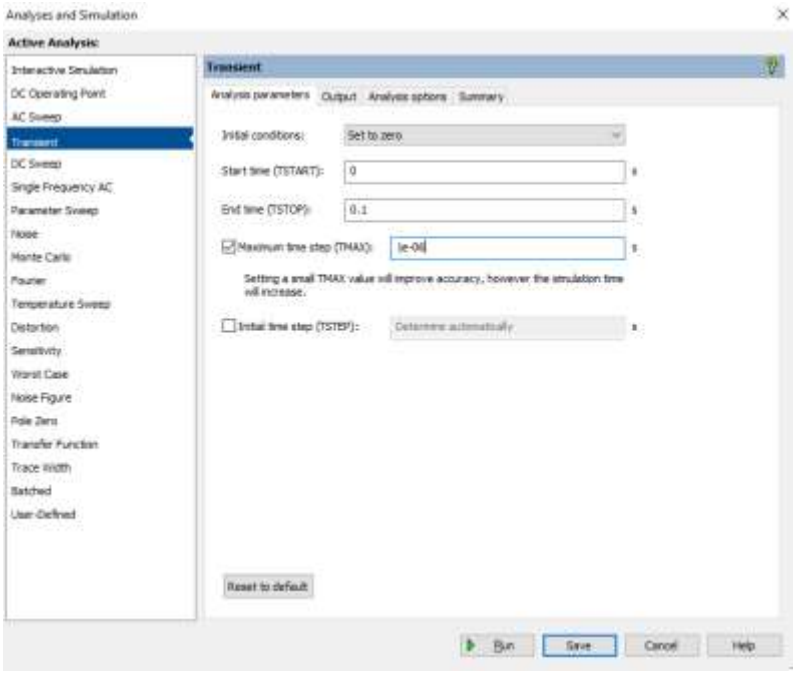


Figure 7

6. Set your output Vc > V2 (depend on net voltage value) > Run.

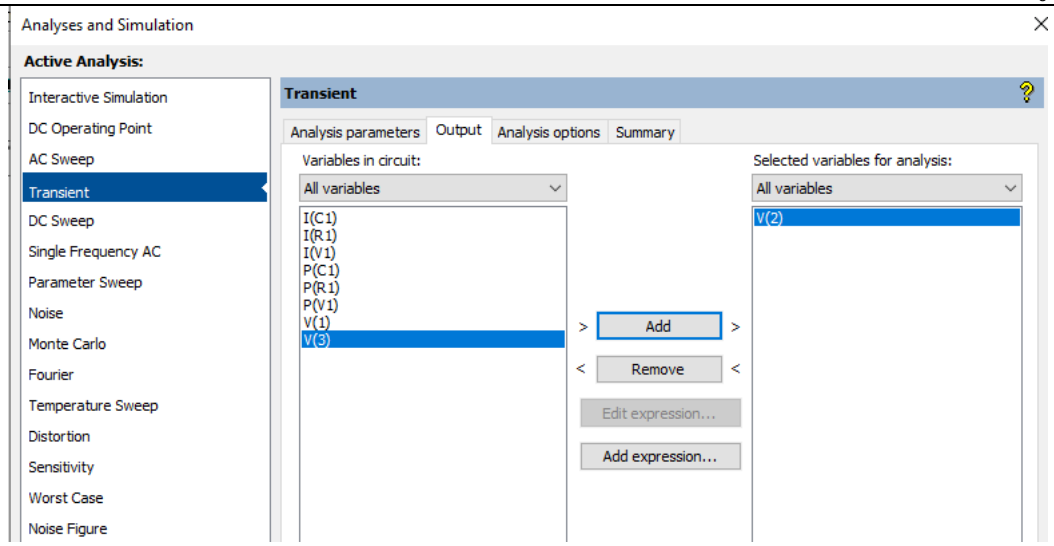


Figure 8

7. Record the V_C voltage waveform and measure the value of the value of τ, 2τ, 3τ, 4τ and 5τ by using cursor in the transient graph.
8. Record your measurement in Table 1.
9. Repeat step 5 until 7 for current flow through capacitor, Ic.
10. Record your measurement in Table 2.
11. Interchange the position of R and C. Repeat step 3 until 7. Record your measurement in Table 3.
12. Plot all the graph response of Vs, Vc and Vr.

RESULT

Table 1: Voltage for charging capacitor, Vc

Parameter	$V_{c(measured)}$	$V_{c(calculated)}$
$\tau =$		
$2\tau =$		
$3\tau =$		
$4\tau =$		
$5\tau =$		

Table 2: Current flow through the circuit, I

Parameter	$I_{(measured)}$	$I_{(calculated)}$
$\tau =$		
$2\tau =$		
$3\tau =$		
$4\tau =$		
$5\tau =$		

Table 3: Voltage for discharging capacitor, Vr

Parameter	$V_{c(measured)}$	$V_{c(calculated)}$
$\tau =$		
$2\tau =$		
$3\tau =$		
$4\tau =$		
$5\tau =$		

PART B:

When a series RL circuit is connected across a supply, voltage and current transients occur until the current attains a steady-state condition. Consider the circuit figure 1, where R represents the coil's resistance or an external resistance. When the switched is closed, current begins to flow into the inductance. The rate of change of current, roc i, will be greatest when the switched is closed. Current and voltage transients will be produced until the current reaches a steady-state level of V/R, at which time the coil effect will be negligible.

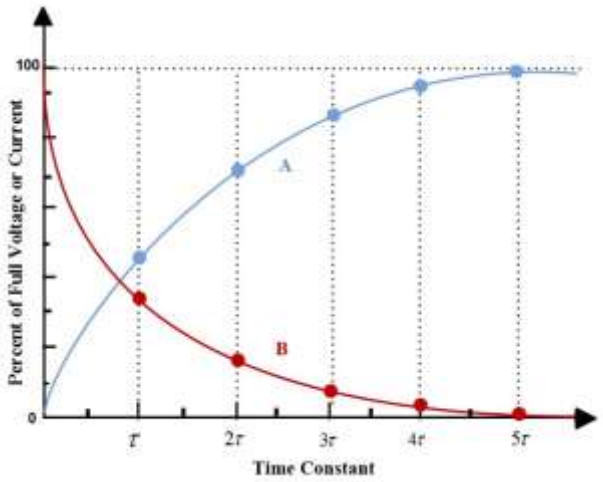


Figure 9: Universal Time Constant Curves for RC and RL circuits

The X-axis represents time constants, and the Y-axis represents a percentage of full current or voltage. For RL circuits, curves A and B represents I_L and V_L respectively. For RC circuits, curves A and B represents V_C and I_C respectively. For both cases, the rise or fall of the curve changes by 63 % in one time constant.

The time required for the transient current to reach 63.2 % of its maximum value can be calculated by the following equation:

$$time\ constant, \tau = \frac{L}{R}\ (s)$$

The time constant τ also represents the time required for the steady-state current to drop 63.2 % when the inductive circuit is opened.

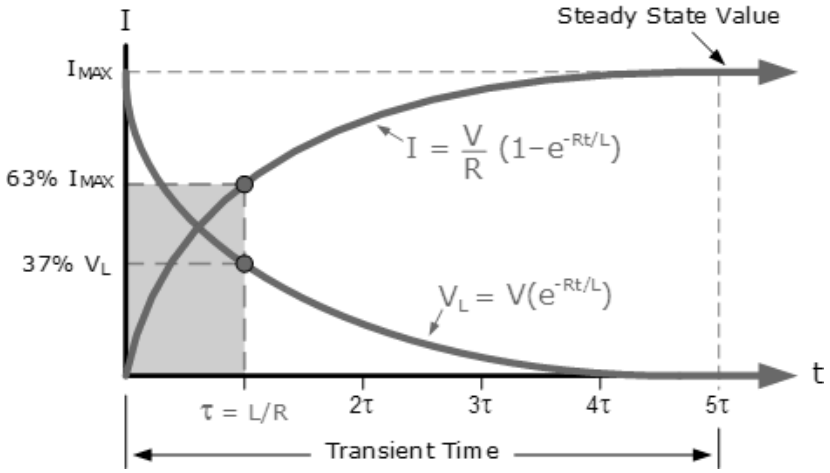


Figure 10

EQUIPMENT AND COMPONENT

1. Step Voltage, 10V
2. Resistor, 20 Ω
3. Inductor, 200mH

PROCEDURE

Circuit diagram

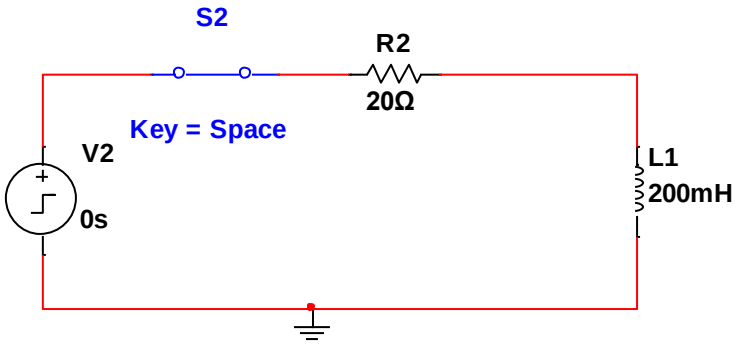


Figure 11

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM106_Experiment 2_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 13 to construct your circuit.
4. Wire up the circuit as shown in Figure 11 where, R = 20 Ω and L = 200mH.
5. Set step input voltage to 10V. Initial level = 0V > Final level =10V > step time = 0s > output rise/fall time = 0.1n s.

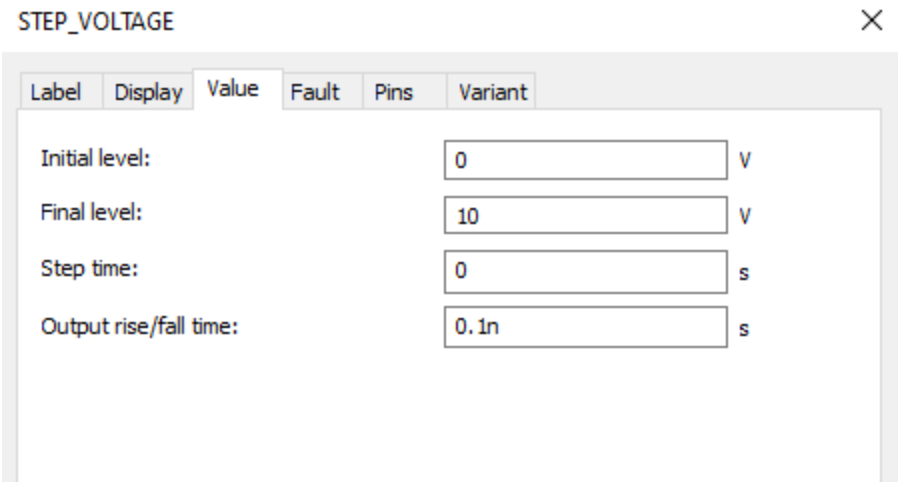


Figure 12

6. Set Analyses and simulation > Transient > Initial condition = set to zero > start time = 0s > end time = 0.1s > maximum time step = 1e-06 s.

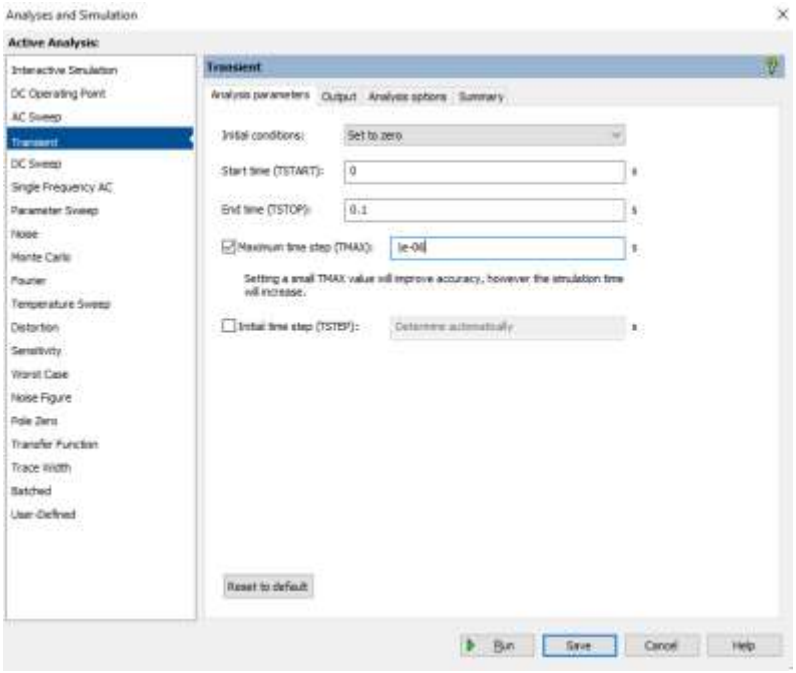


Figure 13

7. Set your output V_L > V1 (depends on net names) > Run.

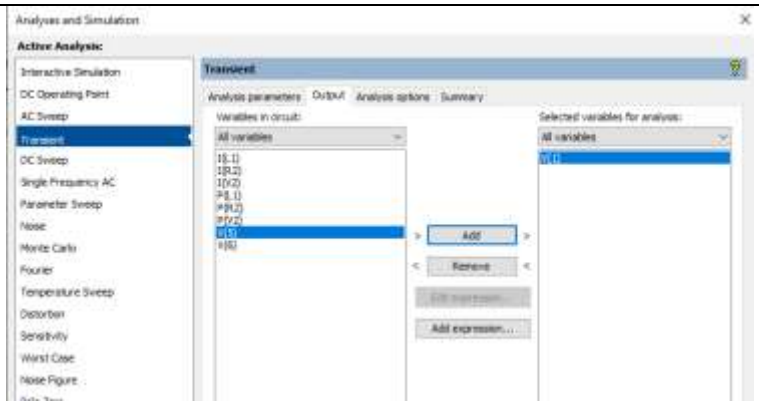


Figure 14

8. Set your output I_L > I(L1) > Run.

9. Record the V_L voltage waveform and measure the value of the value of τ, 2τ, 3τ, 4τ and 5τ by using cursor in the transient graph.
10. Record your measurement in Table 4.
11. Repeat step 5 until 7 for current flow through Inductor, I_L.
12. Record your measurement in Table 5.
13. Interchange the position of R and L. Repeat step 3 until 7. Record your measurement in Table 6.
14. Plot all the graph response of Vs, V_L and V_R.

RESULT

Table 4: Voltage for charging inductor, V_L

Parameter	$V_{c(measured)}$	$V_{c(calculated)}$
$\tau =$		
$2\tau =$		
$3\tau =$		
$4\tau =$		
$5\tau =$		

Table 5: Current flow through the circuit, I

Parameter	$I_{(measured)}$	$I_{(calculated)}$
$\tau =$		
$2\tau =$		
$3\tau =$		
$4\tau =$		
$5\tau =$		

Table 6: Voltage for discharging inductor, Vr

Parameter	$V_{c(measured)}$	$V_{c(calculated)}$
$\tau =$		
$2\tau =$		
$3\tau =$		
$4\tau =$		
$5\tau =$		

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Show your working calculation for Table 1 until Table 6.